

MICROSOFT AZURE

Incremental Data Load Operation (Watermark-based)

Enabled Hierarchical namespace under Advanced settings to make it **Data Lake Gen2**, then clicked Review + Create.

Microsoft Azure Upgrade Search resources, services, and docs (G+/)

Home > Resource Manager | Resource groups > BENCY_DE > Marketplace

Create a storage account

Basics **Advanced** Networking Data protection Security Encryption Tags Review + create

Azure Blob Storage

These are settings which apply to only Azure Blob storage

Enable hierarchical namespace ☒

Enable SFTP ☐

Enable network file system v3 ☐

Allow cross-tenant replication ☐

Cross-tenant replication and hierarchical namespace cannot be enabled simultaneously.

Access tier

☒ Hot
Optimized for frequently accessed data and everyday usage scenarios

☐ Cool
Optimized for infrequently accessed data and backup scenarios

☐ Cold
Optimized for rarely accessed data and backup scenarios

Azure Files

Previous Next **Review + create** Give feedback

Data Factory and Storage account were created successfully.

Microsoft Azure Upgrade Search resources, services, and docs (G+/)

Home > CreateDataFactory-20260828130112 | Overview

BENCY_DE Resource group

Generate Bicep code to duplicate this resource group. Export resource groups using Bicep or Terraform How to manage changes with deployment tools?

Search Create Manage view Delete resource group Refresh Export to CSV Open query Assign tags Move Delete Group by none

Overview Activity log Access control (IAM) Tags Resource visualizer Events Settings Cost Management Monitoring Automation Help

Essentials JSON View

Resources Recommendations

Filter for any field... Type equals all Location equals all Add filter

Name	Type	Location
bencydf	Data factory (V2)	East US
bencyst	Storage account	East US

Showing 1 - 2 of 2. Display count: auto Give feedback

Add or remove favorites by pressing Ctrl+shift+F

Entered server name, selected Location chose SQL authentication, and set admin login credentials

Microsoft Azure | Upgrade | Search resources, services, and docs (G+)

Home > BENCY_DE > Marketplace > Create SQL Database

Create SQL Database Server

Microsoft

Server name * ☒ .database.windows.net

Location *

Authentication

☒ Azure Active Directory (Azure AD) is now Microsoft Entra ID. [Learn more](#)

Select your preferred authentication methods for accessing this server. We recommend using only Microsoft Entra authentication. [Learn more](#) or using an existing Microsoft Entra user, group, or application as Microsoft Entra admin. [Learn more](#)

Authentication method

☐ Use Microsoft Entra-only authentication

☐ Use both SQL and Microsoft Entra authentication

☒ Use SQL authentication

Server admin login * ☒

Password * ☒

Confirm password * ☒

OK [Feedback](#)

Set Hardware to 16 GB data size, then clicked Apply

Microsoft Azure | Upgrade | Search resources, services, and docs (G+)

Home > BENCY_DE > Marketplace > SQL Database > Create SQL Database

Configure

[Feedback](#)

Compute Hardware

Select the hardware configuration based on your workload requirements. Availability of compute optimized, memory optimized, and confidential computing hardware depends on the region, service tier, and compute tier.

Hardware Configuration

Standard-series (Gen5)
up to 128 vCores, up to 625 GB memory
[Change configuration](#)

ESTIMATED COST / MONTH: 413.28 USD

Save money

Already have a SQL Server License? Save with a license you already own with Azure Hybrid Benefit. Actual savings may vary based on region and performance tier. [Learn more](#)

☐ Yes ☒ No

vCores

[Compare vCore options](#)

2

Data max size (GB)

16

4.8 GB LOG SPACE ALLOCATED

Would you like to make this database zone redundant? ☐ Yes ☒ No

Apply

Checked the resource group and confirmed all three resources were

Microsoft Azure | Upgrade | Search resources, services, and docs (G+)

Home > Resource Manager | Resource groups

BENCY_DE
Resource group

[Export resource groups using Bicep or Terraform](#) [Generate Bicep code to duplicate this resource group](#) [Help me generate Terraform for this resource group configuration](#)

[+ Create](#) [Manage view](#) [Delete resource group](#) [Refresh](#) [Export to CSV](#) [Open query](#) [Assign tags](#) [Move](#) [Delete](#) [Group by none](#)

Overview

[Activity log](#) [Access control \(IAM\)](#) [Tags](#) [Resource visualizer](#) [Events](#) [Settings](#) [Cost Management](#) [Monitoring](#) [Automation](#) [Help](#)

Resources [Recommendations](#)

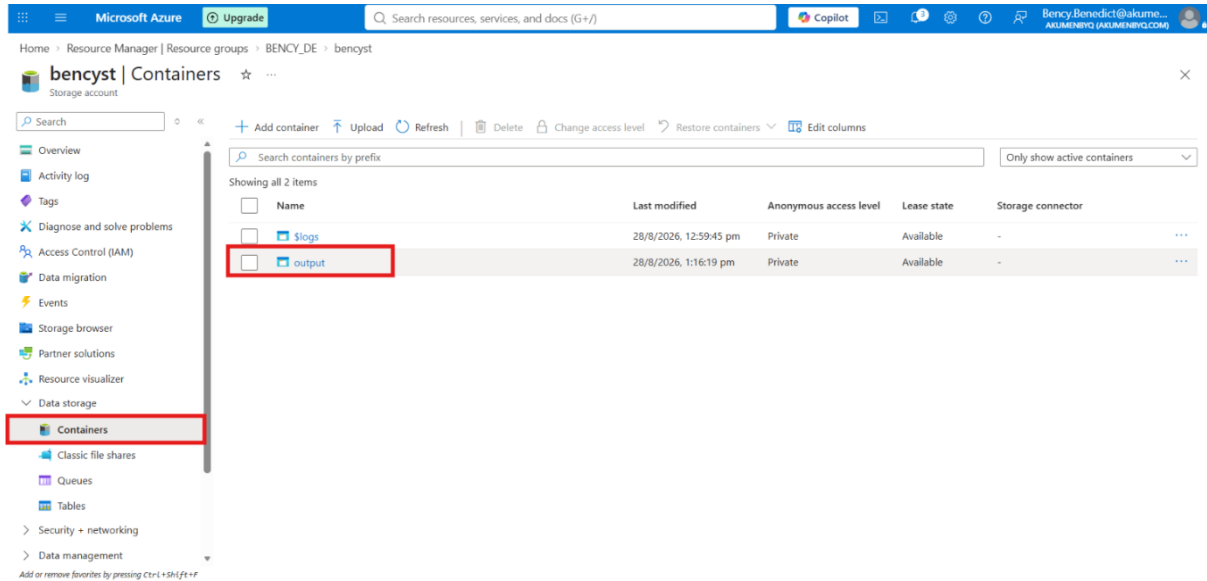
[Type equals all](#) [Location equals all](#) [+ Add filter](#)

☐	Name ↑	Type	Location
☐	bencydf	Data factory (V2)	East US
☐	bencyst	Storage account	East US
☐	sqlserver790	SQL server	Southeast Asia

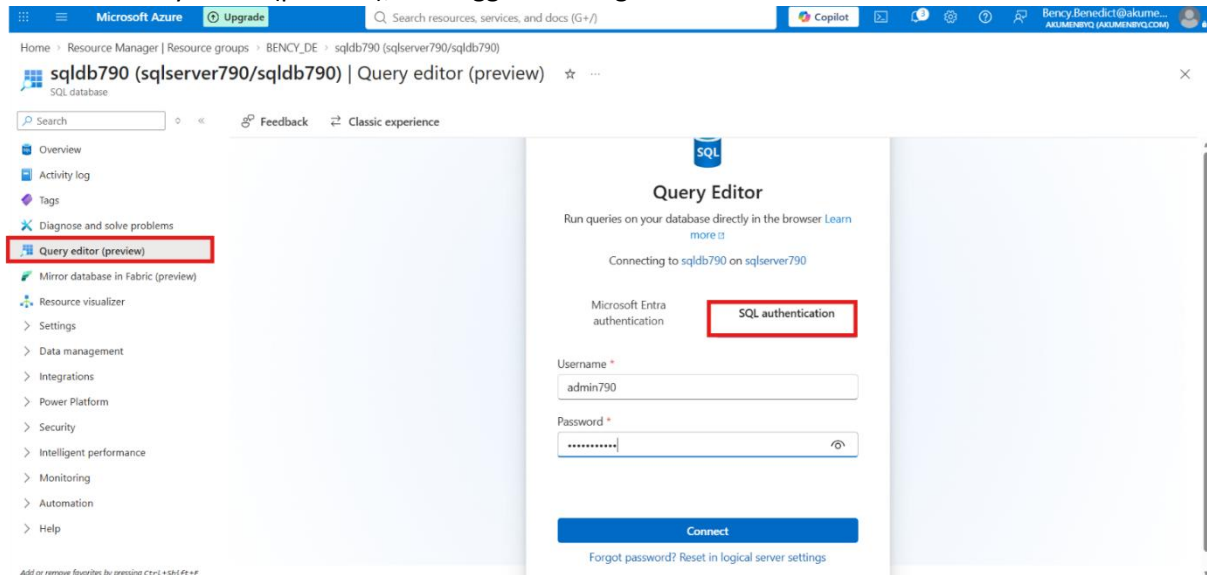
Showing 1 - 3 of 3. Display count:

[Give feedback](#)

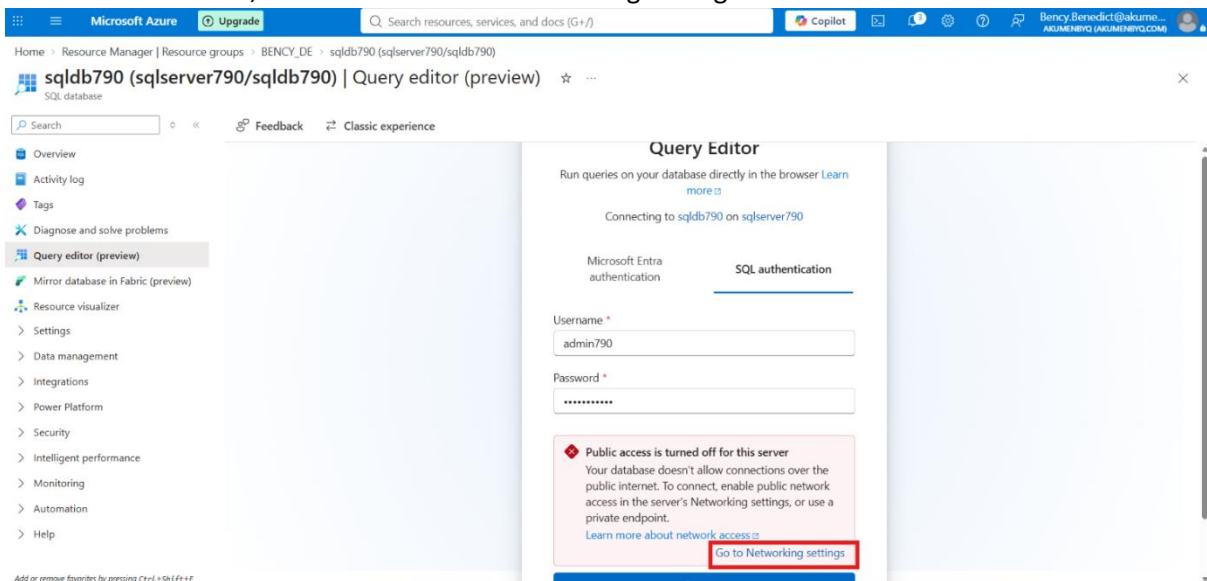
Output container created inside the storage account



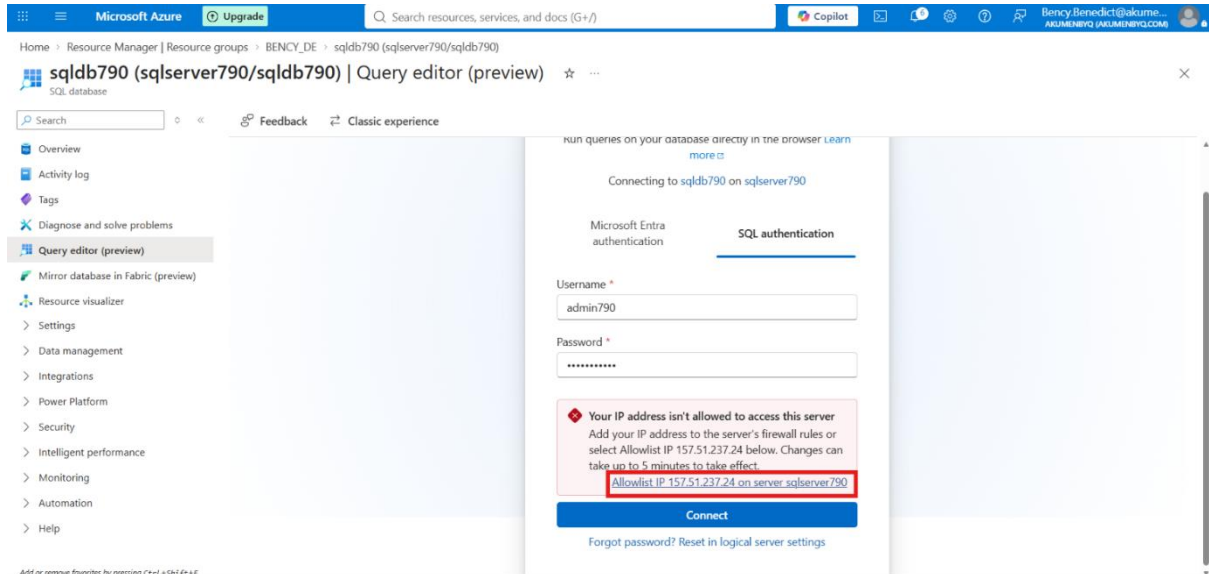
Selected Query editor (preview), and logged in using SQL authentication.



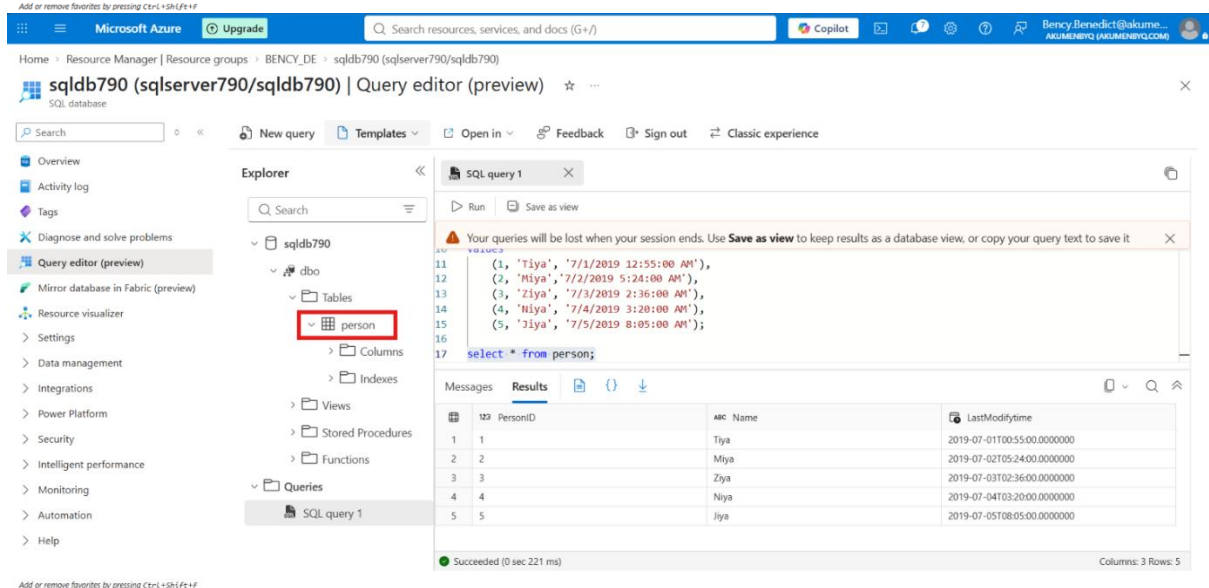
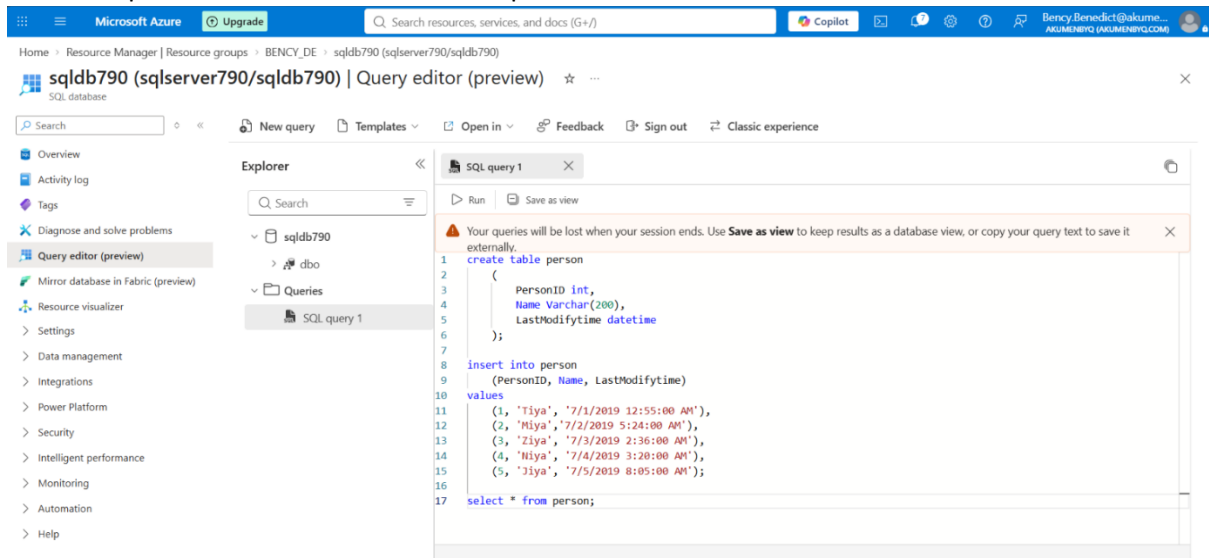
Public access was off, so enabled it under Networking settings.



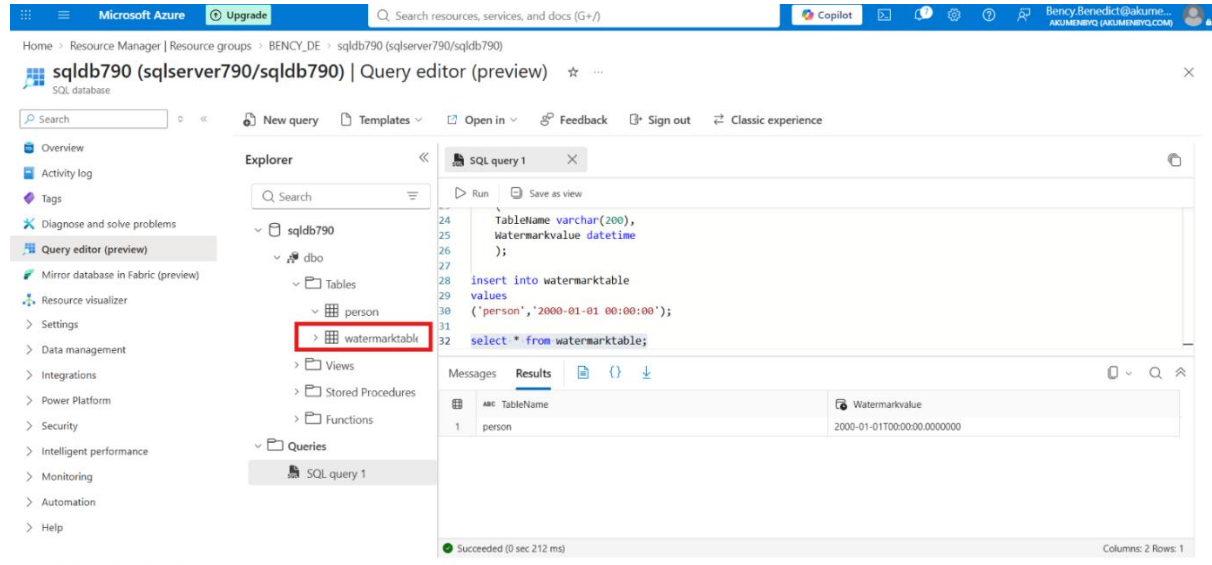
The client IP wasn't allowed, so clicked "Allowlist IP" to fix it.



Created a person table and inserted 5 sample rows



Created a watermarktable and inserted a starting row for person with an initial watermark value



Microsoft Azure | Upgrade | Search resources, services, and docs (G+)

Home > Resource Manager | Resource groups > BENCY_DE > sqldb790 (sqlserver790/sqldb790)

sqldb790 (sqlserver790/sqldb790) | Query editor (preview) ☆

Search

New query | Templates | Open in | Feedback | Sign out | Classic experience

Overview | Activity log | Tags | Diagnose and solve problems | Query editor (preview) | Mirror database in Fabric (preview) | Resource visualizer | Settings | Data management | Integrations | Power Platform | Security | Intelligent performance | Monitoring | Automation | Help

Explorer

Search

sqldb790

dbo

Tables

person

watermarktable

Views

Stored Procedures

Functions

Queries

SQL query 1

SQL query 1

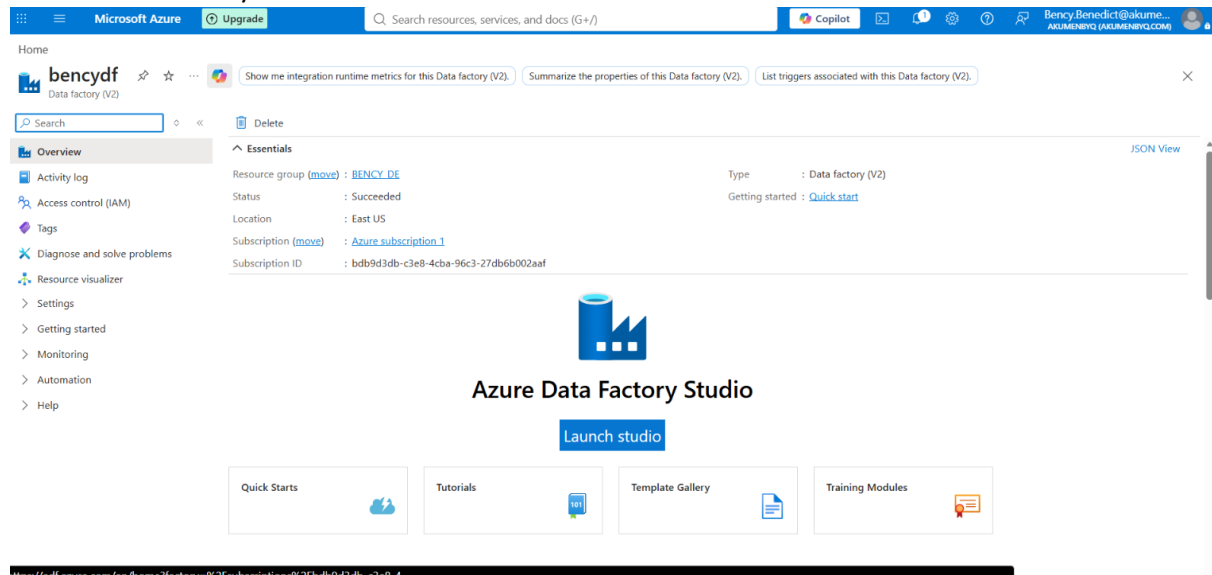
```
24 \
25 \
26 \
27 \
28 insert into watermarktable
29 values
30 ('person', '2000-01-01 00:00:00');
31
32 select * from watermarktable;
```

Messages | Results

Table Name	Watermarkvalue
person	2000-01-01T00:00:00.0000000

Succeeded (0 sec 212 ms) Columns: 2 Rows: 1

Launch Data Factory



Microsoft Azure | Upgrade | Search resources, services, and docs (G+)

Home

bencydf Data factory (V2) ☆

Show me integration runtime metrics for this Data factory (V2). | Summarize the properties of this Data factory (V2). | List triggers associated with this Data factory (V2).

Search

Delete

Overview | Activity log | Access control (IAM) | Tags | Diagnose and solve problems | Resource visualizer | Settings | Getting started | Monitoring | Automation | Help

Essentials

Resource group (move) : BENCY_DE | Type : Data factory (V2) | Status : Succeeded | Getting started : Quick start | Location : East US | Subscription (move) : Azure subscription 1 | Subscription ID : bdb9d3db-c3e8-4c8a-96c3-27db6b002aaf

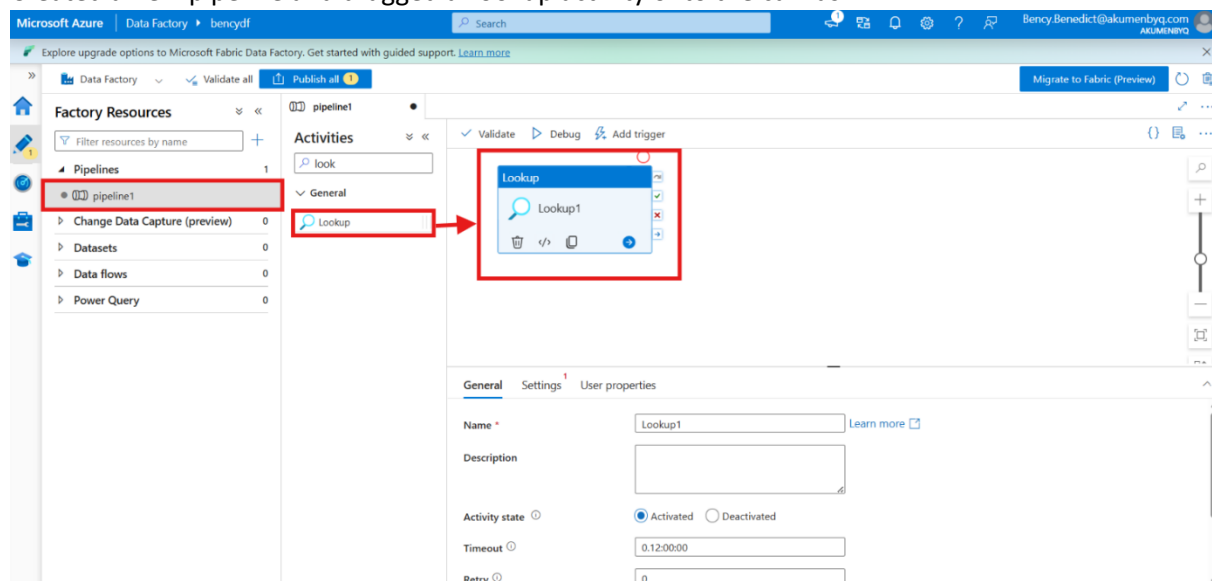
Azure Data Factory Studio

Launch studio

Quick Starts | Tutorials | Template Gallery | Training Modules

<https://adf.azure.com/en/home?factory=%2Fsubscriptions%2Fbdb9d3db-c3e8-4c8a-96c3-27db6b002aaf>

Created a new pipeline and dragged a Lookup activity onto the canvas



Microsoft Azure | Data Factory | bencydf | Search

Explore upgrade options to Microsoft Fabric Data Factory. Get started with guided support. [Learn more](#)

Data Factory | Validate all | Publish all | Migrate to Fabric (Preview)

Factory Resources

Filter resources by name

Pipelines

pipeline1

Change Data Capture (preview) 0

Datasets 0

Data flows 0

Power Query 0

Activities

look

Lookup

Lookup1

General | Settings | User properties

Name *

Lookup1

Description

Activity state

Activated

Deactivated

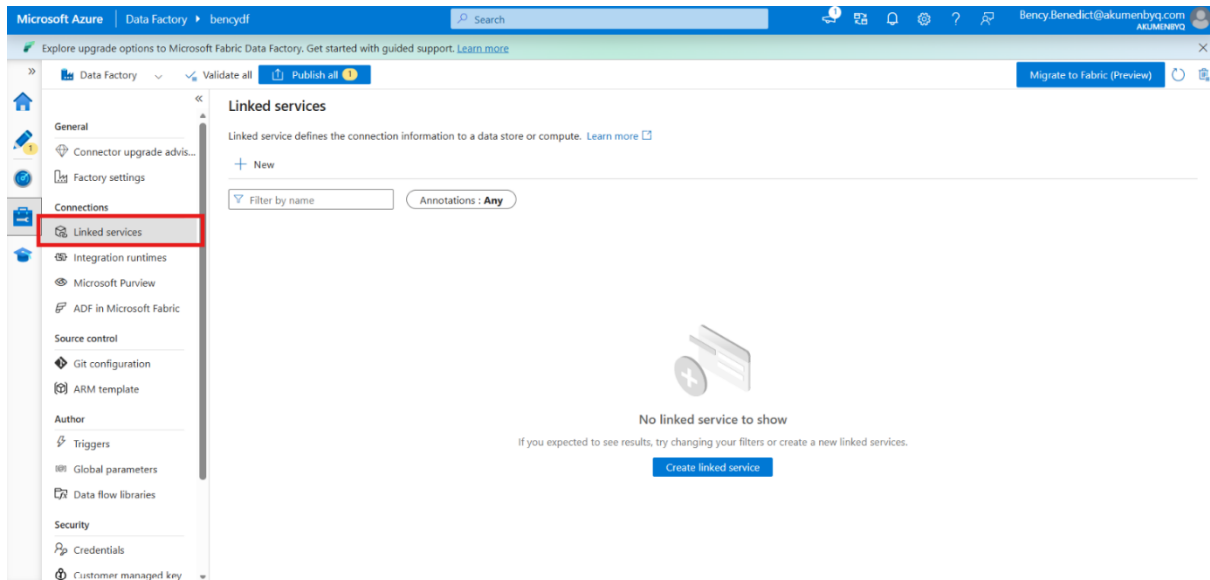
Timeout

0.12:00:00

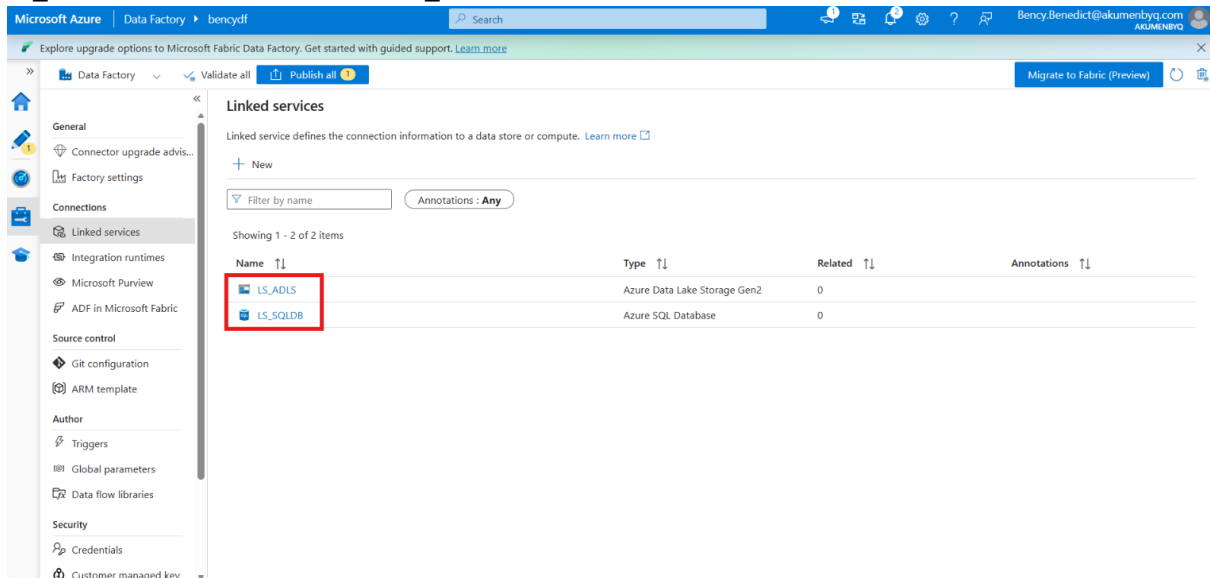
Retry

0

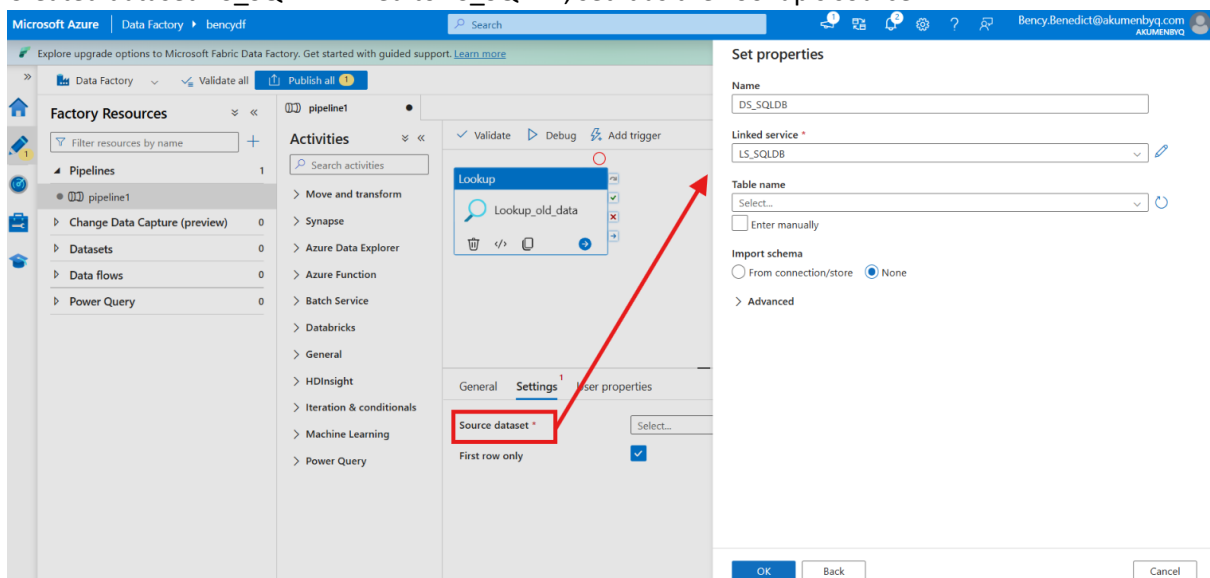
Created two Linked Services



LS_ADLS for Data Lake Gen2 and LS_SQLDB for Azure SQL Database.



Created dataset DS_SQLDB linked to LS_SQLDB, set it as the Lookup's source



Entered query and previewed the data to confirm

The screenshot shows the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane lists 'pipeline1'. The 'Activities' pane shows a 'Lookup' activity named 'Lookup_old_data'. The 'Settings' tab for this activity is selected, showing 'Source dataset' as 'DS_SQLDB', 'First row only' checked, and 'Use query' selected. The query entered is: `select * from watermarktable where TableName = 'person'`. A red box highlights the 'Query' section. Below the settings, the 'Preview data' window is open, showing a table with two columns: 'TableName' and 'Watermarkvalue'. The table contains one row: 'person' and '2000-01-01T00:00:00'. A red arrow points from the 'Preview data' button in the bottom right of the settings pane to the 'Preview data' window.

TableName	Watermarkvalue
person	2000-01-01T00:00:00

New Lookup activity dragged onto the canvas and entered the query

The screenshot shows the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane lists 'pipeline1'. The 'Activities' pane shows two 'Lookup' activities: 'Lookup_old_data' and 'Lookup_new_data'. The 'Settings' tab for 'Lookup_new_data' is selected, showing 'Source dataset' as 'DS_SQLDB', 'First row only' checked, and 'Use query' selected. The query entered is: `select max(LastModifytime) as NewWatermarkvalue from person`. A red box highlights the 'Query' section.

TableName	Watermarkvalue
person	2000-01-01T00:00:00

Previewed the data to confirm

Microsoft Azure | Data Factory | bencydf

Explore upgrade options to Microsoft Fabric Data Factory. Get started with guided support. [Learn more](#)

Factory Resources

- Pipelines 1
 - pipeline1
- Change Data Capture (preview) 0
- Datasets 1
- Data flows 0
- Power Query 0

Preview data

Linked service: LS_SQLDB

Object:

	NewWatermarkvalue
1	2019-07-05T08:05:00

Query timeout (minutes) 120

New [Preview data](#) [Learn more](#)

CopyData activity is selected and dragged into panel

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Factory Resources

- Pipelines 1
 - pipeline1
- Change Data Capture (preview) 0
- Datasets 1
- Data flows 0
- Power Query 0

Activities

- cop
- Move and transform
 - Copy data

Lookup

- Lookup_old_data
- Lookup_new_data

Copy data

Copy data1

General

Name * Copy data1 [Learn more](#)

Description

Activity state ☒ Activated ☐ Deactivated

Timeout 0.12:00:00

Retry 0

Connect the pipeline activities

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Factory Resources

- Pipelines 1
 - pipeline1
- Change Data Capture (preview) 0
- Datasets 1
- Data flows 0
- Power Query 0

Activities

- cop
- Move and transform
 - Copy data

Lookup

- Lookup_old_data
- Lookup_new_data

Copy data

Copy data1

Parameters

Variables

Settings

Output

+ New

Entered the query in source

The screenshot shows the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane lists 'Pipelines', 'Datasets', 'Data flows', and 'Power Query'. The 'Activities' pane shows 'Lookup' and 'Copy data'. The main canvas displays a pipeline with two 'Lookup' activities ('Lookup_old_data' and 'Lookup_new_data') connected to a 'Copy data' activity. The 'Copy data' activity is selected, and the 'Source' tab is active. The 'Source dataset' is 'DS_SQLDB'. The 'Use query' option is selected, and the query is 'SELECT * FROM person WHERE Last...'. The 'Query timeout (minutes)' is set to 120. The 'Partition option' is set to 'None'.

The screenshot shows the Microsoft Azure Data Factory interface with the 'Pipeline expression builder' dialog open. The dialog contains a text area with the following SQL query:

```
SELECT *
FROM person
WHERE LastModifytime > '{activity('Lookup_old_data').output.firstRow.Watermarkvalue}'
AND LastModifytime <= '{activity('Lookup_new_data').output.firstRow.NewWatermarkvalue}'
```

Below the text area, there are tabs for 'Activity outputs', 'Parameters', 'System variables', 'Functions', and 'Variables'. The 'Activity outputs' tab is selected, showing a list of activity outputs with their first row data.

Stored Procedure activity is dragged into the panel

The screenshot shows the Microsoft Azure Data Factory interface. The 'Factory Resources' pane lists 'Pipelines', 'Datasets', 'Data flows', and 'Power Query'. The 'Activities' pane shows 'Lookup' and 'Copy data'. The main canvas displays a pipeline with two 'Lookup' activities ('Lookup_old_data' and 'Lookup_new_data') connected to a 'Copy data' activity. A 'Stored procedure' activity is being dragged from the 'Activities' pane into the pipeline. The 'Stored procedure' activity is highlighted with a red box. The 'General' tab is active, showing the 'Name' as 'Stored procedure1', 'Description' as an empty field, 'Activity state' as 'Activated', 'Timeout' as '0:12:00:00', and 'Retry' as '0'.

Set Linked service, stored procedure name

The screenshot shows the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane lists 'Pipelines' (1), 'Datasets' (2), 'Data flows' (0), and 'Power Query' (0). The 'Activities' pane shows 'Lookup' and 'Stored procedure'. The main canvas displays a pipeline diagram with two 'Lookup' activities ('Lookup_old_data' and 'Lookup_new_data') feeding into a 'Copy data' activity ('Copy data1'), which then feeds into a 'Stored procedure' activity ('Stored procedure1'). The 'Stored procedure' activity is selected, and its 'Settings' tab is active. A red box highlights the 'Linked service' dropdown set to 'LS_SQLDB' and the 'Stored procedure name' dropdown set to '[dbo].[write_watermarktable]'. Below these, there is a 'Test connection' button showing 'Connection successful' and a 'Refresh' button. The 'Stored procedure parameters' section is also visible.

Import its parameters

The screenshot shows the 'Stored procedure parameters' section of the 'Stored procedure' activity. A red box highlights the 'Import' button. Below it, a table lists parameters to be imported:

Name	Type
LastModifiedtime	DateTime
TableName	String

The 'Import' button is highlighted with a red box. The 'Stored procedure parameters' section is expanded, showing the 'Import' button and the table of parameters to be imported.

Entered the values

The screenshot shows the 'Stored procedure parameters' section of the 'Stored procedure' activity. A red box highlights the 'Import' button and the table of parameters to be imported, with values entered for 'LastModifiedtime' and 'TableName'.

Name	Type	Value
LastModifiedtime	DateTime	@activity('Lookup_new_data').output...
TableName	String	@activity('Lookup_old_data').output.firs...

Ran Validate and Debug

Microsoft Azure | Data Factory | bencydf

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Factory Resources

- Pipelines: 1
 - pipeline1
- Datasets: 2
 - DS_ADLS
 - DS_SQLDB
- Data flows: 0
- Power Query: 0

Activities

- store
- Stored procedure

Lookup

Copy data

Stored procedure

Parameters Variables Settings Output

All status

Showing 1 - 4 of 4 items

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime
Stored procedure1	✓ Succeeded	Stored procedu	8/31/2026, 1:17:26 PM	18s	AutoResolveIntegrationRuntime (Centri
Copy data1	✓ Succeeded	Copy data	8/31/2026, 1:17:06 PM	19s	AutoResolveIntegrationRuntime (East U
Lookup_old_data	✓ Succeeded	Lookup	8/31/2026, 1:16:54 PM	10s	AutoResolveIntegrationRuntime (Centri
Lookup_new_data	✓ Succeeded	Lookup	8/31/2026, 1:16:54 PM	9s	AutoResolveIntegrationRuntime (Centri

Confirmed 5 rows were copied

Microsoft Azure | Data Factory | bencydf

Details Refresh

Learn more on copy performance details from here.

Activity run id: 322219be-b6aa-4af9-8612-929d3d0487aa

Succeeded

Azure SQL Database Region: Southeast Asia

Azure Data Lake Storage Gen2

Data read: 100 bytes
Rows read: 5
Peak connections: 1

Data written: 220 bytes
Files written: 1
Rows written: 5
Peak connections: 1

Copy duration: 00:00:15
Throughput: 14 bytes/s

Azure SQL Database → Azure Data Lake Storage Gen2

Start time: 8/28/2026, 1:50:39 PM
Used DUs: 4
Used parallel copies: 1

Duration: 00:00:15

Details	Working duration	Total duration
Queue	00:00:00	00:00:06
Transfer	00:00:00	00:00:07

Person records appeared correctly in the output

Microsoft Azure | Upgrade

Search resources, services, and docs (G+/)

Home > Resource Manager | Resource groups > BENCY_DE

output

Authentication method

Showing all 3 items

data_bb1e179f-cec0-4fbb-b9e0-a99b8a667d4f_89cb7455-c9d3-4fc8-8c92-705bf53f0a99.csv

Overview Versions Edit Generate SAS

```
1 PersonID,Name,LastModifyTime
2 1,"Tiya",2019-07-01 00:55:00.0000000
3 2,"Miya",2019-07-02 05:24:00.0000000
4 3,"Ziya",2019-07-03 02:36:00.0000000
5 4,"Niya",2019-07-04 03:20:00.0000000
6 5,"Iiya",2019-07-05 08:05:00.0000000
7
```

Re-ran the pipeline — this time 0 bytes and 0 rows were affected

The screenshot shows the 'Details' view of a pipeline run in Microsoft Azure Data Factory. The pipeline is named 'bencydl'. The activity is a 'Copy' operation from 'Azure SQL Database' (Region: Central India) to 'Azure Data Lake Storage Gen2' (Region: East US). The status is 'Succeeded'. The activity run ID is 'f13570bf-069c-41e4-8b85-f016c17f8930'.

Copy Details:

- Data read: 0 byte
- Rows read: 0
- Peak connections: 1
- Data written: 30 bytes
- Files written: 1
- Rows written: 0
- Peak connections: 1

Copy duration: 00:00:14
Throughput: 0 byte/s

Azure SQL Database → Azure Data Lake Storage Gen2

Start time: 8/31/2026, 1:17:07 PM
Used DIUs: 4
Used parallel copies: 1
Duration: 00:00:14

Details	Working duration	Total duration
Queue	00:00:03	00:00:06
Transfer	Time to first byte: 00:00:03 Reading from source: 00:00:00 Writing to sink: 00:00:00	00:00:06

Confirmed the second output csv had only the header row, validating the incremental watermark logic worked correctly.

The screenshot shows the 'Overview' view of a blob file in Microsoft Azure. The blob is named 'data_c63c1bc5-32d2-4fa5-94f8-d607d74b200e_cf6039a5-2255-4bf8-bcc3-31ad5011bd7c.csv'. The 'Edit' button is highlighted. The file content is displayed as a table with two columns: 'PersonID' and 'Name, LastModifiedByTime'.

PersonID	Name, LastModifiedByTime

The file is a CSV file, and the 'Preview' button is visible at the bottom.